An integrated computational framework for diversity-sensitive personalized medicine

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Abstract

Diversity in biological, social, and environmental factors plays a central role in shaping brain health and disease. Distinct brain disorders frequently exhibit overlapping clinical phenotypes, despite arising from heterogeneous biological and contextual mechanisms. This convergence challenges conventional, population-averaged approaches, which often fail to capture interindividual variability and lead to limited reproducibility, weak translational potential, and inadequate tools for individual-level characterization. To address these gaps, we propose an integrative computational framework that unites normative models of brain aging ("brain clocks"), high-order interactions, and whole-brain modeling. Brain clocks estimate individualized brain health scores by comparing observed brain features to normative age-based trajectories. Brain high-order interactions capture functional dependencies beyond pairwise connectivity, offering sensitive biomarkers that reflect system-level diversity in aging and neurodegeneration. Whole-brain modeling uses theory-based simulations of individual brain dynamics, supporting the inference of latent mechanisms and the evaluation of targeted perturbations *in silico*. Together, these approaches form a synergistic framework: normative models provide personalized baselines, high-order interactions enhance sensitivity to complex alterations, and whole-brain simulations enable causal insight and guide potential interventions. By embedding interindividual variability and contextual diversity into each computational layer, this framework moves the field toward precision neuroscience, where assessment, understanding, and treatment are tailored to the individuals' unique biological and social profiles.

Keywords: whole-brain modeling, personalized medicine, high-order interactions, brain health, dementia, diversity, global south

Introduction

Diverse genetic backgrounds and socio-environmental exposomes –spanning income inequality, chronic stress, nutritional deficits, and exposure to violence– shape brain development and aging worldwide[1-3]. This influence is particularly pronounced in the Global South, where health systems are under-resourced and research participation remains low[4,5]. These regions are characterized by high levels of genetic diversity and distinct social and environmental exposomes[5,6], that are known to impact brain development and aging[1-3]. Neuropsychiatric and other brain disorders are major contributors to global disability and disease burden, particularly in aging populations[7]. Among brain disorders, dementias alone already afflict over 55 million people globally, with projections surpassing 139 million by 2050[8], and impose an even greater burden in low- and middle-income regions, worsening disparities in care and outcomes. Existing frameworks, however, lack integrative, mechanistic insight, yielding results that are often irreproducible and poorly generalize across populations[9,10]. In response, integrative, multimodal computational approaches offer a promising route: by combining advanced analysis of multimodal large-scale neuroimaging data with statistical and biophysical modelling, such methods can more faithfully characterize healthy aging trajectories, reveal context-specific deviations, and provide underlying biological mechanisms[11,12].

There are significant barriers to implementing computational approaches in personalized medicine. Brain-derived data are often noisy, short in duration, and collected using heterogeneous protocols, limiting reproducibility and generalizability[13,14]. These issues are especially pronounced in under-resourced regions, where access to advanced equipment and standardized methods is limited[15,16]. However, beyond data limitations, meaningful clinical applications require frameworks that capture population-specific variability. Indeed, normative models –used to identify deviations from typical brain development or aging[17]– often fail to generalize when tested in diverse or underrepresented populations[10]. Their lack of sensitivity to biological and environmental factors limits their application in global health. Computational neuroscience offers powerful tools to address these gaps. Advanced tools from network science, dynamical systems, information theory, machine learning, and modeling techniques can identify population-sensitive biomarkers and elucidate underlying mechanisms of brain dysfunction[2,3,18-20]. These approaches not only enable the detection of disease-relevant deviations but also provide a window into the mechanistic processes shaped by genetics and the exposome[19,21-24], offering an integrative framework that can account for diverse populations. These methods can be remotely deployed and scaled across cohorts, supporting cost-efficient, personalized, context-sensitive brain health modeling.

Here, we outline an integrated computational framework to improve brain health research across diverse populations. This framework includes: (i) brain clocks, which operationalize normative brain aging and estimate deviations at the individual level[25,26]; (ii) brain higher-order interactions (HOIs)[27-31], which capture complex multivariate dependencies in neural systems beyond pairwise functional connectivity; and (iii) biophysical whole-brain models (WBMs)[22,32,33], which simulate mechanistic processes underlying brain dynamics. By leveraging these complementary approaches, we propose a methodological roadmap to identify context-sensitive markers of brain health, aging, and disease, and to inform mechanistic brain function models responsive to population diversity. This synergy may lay the groundwork for a more precision-oriented neuroscience, with broad implications for personalized medicine.

Diversity-sensitive brain clocks as normative models of brain aging

Brain clocks are computational models that estimate an individual's brain age from neuroimaging data, providing a normative score of brain health[17,25,26] (**Figure 1**). Brain clocks start using machine learning or deep learning algorithms –such as support vector regression, Gaussian processes, or graph-based neural networks– trained on large functional magnetic resonance imaging (fMRI), electroencephalography (EEG), or MRI datasets to predict chronological age[34-36] (**Figure 1A**). These models typically rely on preprocessed features like connectivity matrices or HOIs derived from parcellated time series (**Figure 1B**). After hyperparameter tuning and cross-validation, they can be used to predict age from brain-derived data. These models yield a brain age prediction[37], with the brain age gap (BAG) calculated as the difference between predicted and actual age[25,26] (**Figure 1C**).

The key output of these models is the BAGs[25,26,38]. A positive BAG indicates accelerated brain aging, while a negative BAG reflects delayed aging relative to the normative reference[25,26,38]. These individualized scores allow for quantifying deviations from expected brain aging trajectories and have emerged as useful biomarkers for assessing brain health across diverse populations[25]. Because brain clocks are grounded in normative modeling, they support comparative analyses between individuals and groups, offering insight into how genetic, clinical, and environmental factors shape brain aging across the lifespan[17,26]. When paired with multimodal neuroimaging and electrophysiological data, brain clocks represent a scalable, simple, and practical framework for studying brain health in global settings, including basic and clinical research.

Brain clocks have been applied to characterize brain aging in the context of neurological and psychiatric disorders[25,26,37,39-41]. They have shown consistent evidence of accelerated brain aging across conditions such as schizophrenia, major depression, epilepsy, and dementia-related syndromes[26,37,39-41]. For example, brain clocks can be extended by incorporating functional HOIs into models, improving their sensitivity to subtle and distributed alterations in brain dynamics[25]. This has proven especially valuable in characterizing aging deviations across the dementia continuum –including mild cognitive impairment, Alzheimer's disease, and frontotemporal dementia– where increased BAGs were observed, consistent with disease progression[25]. These models based on HOIs have also revealed marked differences between populations from the Global North and South, detecting accelerated brain aging in Latinos versus the Global North participants, which was modulated by biological sex, even after controlling for technical and demographic confounds[25,41]. Importantly, the robustness of these models has been supported by harmonization pipelines[42-45] that reduce variability arising from different recording devices, acquisition protocols, and signal quality[42-45].

Beyond disease classification, brain clocks can be applied to assess the influence of lifestyle and interventions on brain aging[26,37,46-50]. BAGs have been linked to modifiable risk factors such as alcohol consumption, physical activity, and dietary patterns[26]. For example, individuals engaged in creative activities –such as playing musical instruments, dancing, visual arts, or strategy-based video games– exhibit delayed brain aging across several domains[51], through mechanisms related to neural plasticity[52-55]. Furthermore, using a pre/post-learning design, brain clocks can detect short-term brain changes in response to non-pharmacological interventions[51], offering a framework to monitor the direct impact of experiential and cognitive stimulation on brain health. In individuals with schizophrenia, brain clocks have also been used to assess the effects of physical training, revealing delayed brain aging following intervention, with BAG reductions correlating with improvements in clinical symptoms[56]. In this way, brain clocks can provide scientifically based evidence about the effects of modifiable factors and pharmacological therapies on brain health, contributing to delivering more precise information about preventive and tailored strategies for public health policies.

Emerging directions in brain clock research aim to bridge these models with other biological clocks, such as epigenetic, metabolomic, and organ-specific clocks[26,57,58] to build more integrative assessments of biological aging. The integration of brain clock models with HOIs informarkers holds significant potential for advancing our ability to characterize the neural mechanisms underlying aging[25,29,30]. This approach may facilitate earlier detection and enable more personalized monitoring of individuals at elevated risk for age-related cognitive decline[59]. Moreover, advances in low-cost and scalable technologies, such as portable EEG systems, enable the deployment of brain clock models in resource-limited settings, expanding their accessibility, especially in low and middle-income countries. These portable devices, integrated with brain clocks, can lead to the development of real-life health monitoring tools. Another promising avenue is the development of region or network-specific brain clocks that track aging within distinct functional or structural brain systems[60]. These mesoscopic-level models can reveal disease-specific patterns of accelerated aging and may improve the specificity and interpretability of brain health assessments across neurological and psychiatric conditions.

As some technical considerations, despite BAGs can be useful as individual-level normative scores, BAGs can be biased by age-related regression to the mean[61,62]. This requires correction methods like residualization or age-matched designs, though these may artificially inflate model performance if not applied cautiously[63].

Disentangling brain complexity through higher-order interactions

Phenomena such as brain aging and dementia involve a complex interplay of factors and are rarely due to dysfunction within isolated brain regions[64-67]. Over the past two decades, research has shifted from focusing on individual structures to integrative approaches that examine the architecture of interactions among brain regions and neurons[68]. In particular, the study of brain network properties has improved our ability to distinguish between cognitive and pathological states and to identify the mathematical features of network organization that support specific cognitive functions or contribute to disease[69]. Traditionally, these strategies focused mainly on the study of pairwise interactions, such as synapses between two neurons or correlation between two brain areas. While pairwise interactions remain central to studying brain dynamics, growing evidence supports that HOI, involving three or more neurons or brain regions simultaneously, offer deeper insights across scales, from neuronal activity to complex phenomena like social contagion[70].

HOIs can be characterized from different angles. Two of them are based on based on topological data analysis and those grounded in information theory (**Figure 2A**)[71-74]. Topological data analysis provides a framework to explore the higher-order architecture of brain data by identifying geometrical shapes and patterns of interdependencies with specific mathematical properties of interest[74]. One widely used tool in topological data analysis is persistent homology, which focuses on topological features like cycles (e.g., recurring pathways of functional connectivity) or voids (gaps in network structure). This allows researchers to detect robust, multiscale features that traditional pairwise analyses often overlook. Insights from persistent homology can be used to build representations like the homological scaffold, that is, a network where the strength of each connection reflects its involvement in these topological patterns, rather than just direct correlations between regions[75-77].

Persistent homology has been applied to investigate a broad range of cognitive and pathological conditions[78,79]. The study of specific topological features, such as one-dimensional cycles, which represent loops of interconnected brain regions, have garnered particular interest in neuroscience, with ready-to-use toolboxes already accessible[80]. They enable the characterization and prediction of functional alterations associated with psychiatric and neurobiological conditions such as schizophrenia[81], epilepsy[78], attention-deficit/hyperactivity disorder and autism spectrum disorder[82], as well as to describe structural networks in healthy individuals[83,84]. Further, persistent homology showed greater sensibility in characterizing altered states of consciousness than the traditional pairwise approaches[84]. Furthermore, recent research has begun to explore the dynamical properties of homological scaffold, demonstrating that they can improve subject identifiability beyond pairwise static and pairwise approaches, suggesting their potential relevance for personalized medicine[73].

On the side of information theory, HOIs are captured via measuring the total statistical interaction strength (i.e., shared information) among groups of brain regions from their activity recordings (e.g., fMRI or EEG)[27]. Moreover, different types of shared information can be distinguished, specifically, redundancy (information shared across the group of variables) and synergy (information available only when considering the joint state of the whole group of variables)[85]. Multiple metrics and computational pipelines have been introduced to quantify redundancy and synergy[80,86,87], and recent work introduced time-resolved extensions[71,88], allowing for their temporal tracking and revealing dynamic changes in brain coordination that static analyses may miss.

Information theory-based HOIs have been largely used to characterize development and lifespan changes in the brain of healthy populations, characterizing these phenomena in terms of synergy and redundancy. For example, in infants, a recent study identified a developmental trajectory in brain networks characterized by a shift from redundancy- to synergy-dominated neural interactions[89], modulated by environmental enrichment, particularly through the mother-infant connection. Throughout the lifespan, redundancy has been proposed as an informational marker that increases in healthy aging and has been interpreted as a loss of specialization, observed in interactions not only of triplets[90] but across all orders of interaction analyzed[30]. In neurodegeneration, HOIs has been used to provide an accurate multimodal (EEG and fMRI) characterization of two forms of dementia: frontotemporal dementia and Alzheimer's disease[28]. On both modalities, dementia was associated with both high-order hyper and hypoconnectivity, but dominated by the former, consistent with pairwise-based analysis pointing towards dysconnectivity in dementia[64]. Moreover, HOIs revealed specific patterns for different dementia subtypes, providing higher sensitivity compared to classical pairwise analyses[28].

Finally, HOIs can be used to characterize the effects of non-invasive neuromodulation. For example, HOIs have been used to assess the effects of low-intensity transcranial ultrasound stimulation, an emerging technique able to target cortical and deep brain structures, on brain dynamics[91]. Although the effects of neuromodulation remain unknown, HOIs has proven effective in detecting stimulation-induced perturbations that extend beyond the targeted regions through functional networks in both macaques[92] and humans[93]. These effects are target-specific and point to the potential of HOIs for individualized assessment, moving beyond pairwise group-level analyses to enhance the precision and clinical relevance of neuromodulation at a single subject level. Encouragingly, public perception of such non-invasive approaches is generally positive[94], supporting their future clinical adoption.

Altogether, HOIs provides a novel dimension for analyzing brain functional and structural data, enabling a more refined characterization of individual subjects as well as inter-group differences across various cognitive states and health conditions. Recent literature has emphasized the potential of HOI to enhance our ability to interpret complex brain data (**Figure 2B**). In this context, HOIs can be integrated with brain age prediction models, such as brain clocks, to improve the assessment of BAGs[25]. Furthermore, the study of HOIs, if combined with WBMs, offer a powerful framework for exploring the effects of specific mechanisms at multiple interaction levels.

Whole-brain modeling as a clinical test bench for interventions

Statistical and machine learning approaches offer powerful predictive capabilities but remain as “black boxes”, limiting mechanistic insights into brain dynamics and function. WBMs provide a complementary alternative and enable a mechanistic understanding of brain health, disorders, accelerated aging, and population-level diversity[33]. They provide a principled approach to link brain structure, dynamics, and function, and have recently been called “an essential tool for understanding brain dynamics”[95]. To simulate brain dynamics, WBMs leverage multimodal brain data by incorporating neuroimaging and biophysical priors, such as brain anatomical (structural) connectivity, gene and protein expression maps, or atrophy maps. The typical WBM framework involves using the model to reproduce some empirical features of brain dynamics by optimizing free parameters with biological interpretation[33,95]. In this way, it is possible to provide interpretability and a causal framework where empirical patterns of healthy and pathological brain dynamics can be linked to specific neural mechanisms. The goal would be to develop the so-called ‘digital twins’ (virtual brains)[96,97]. Digital twins are not replicas of the brain *per se*, but models able to reproduce brain dynamics and allow testing mechanistic hypotheses can be tested *in silico*[96]. They provide a means to identify, beforehand, subject-specific vulnerabilities and potential treatment targets, which can then be tested in real experimental settings.

WBMs are typically built by integrating three key components (**Figure 3A**)[32,33]: (i) empirical priors, (ii) local dynamics, and (iii) an objective function. First, empirical priors serve as the anatomical and physiological scaffold of the model. These include subject-specific or population-derived data such as structural connectivity (from diffusion MRI), functional parcellations, gene expression gradients, neurotransmitter receptor densities (obtained from positron emission tomography; PET), or disease-specific atrophy maps[98-102]. Second, local dynamics are defined by mathematical models that generate the activity of each brain region. These may range from pure phenomenological models[103-105] to biophysical models[106,107], covering the whole range of neural activity from the neuron level to large-scale brain dynamics[32]. Finally, the model is calibrated to match empirical brain activity by tuning a set of biologically meaningful parameters, such as global coupling, local excitation–inhibition balance, or average firing rates across regions. This process aims to reproduce key brain features, including functional connectivity, power spectra, or dynamic patterns. To guide this fitting, an objective function is used, typically based on similarity measures, for example, the correlation between empirical and simulated functional connectivity

Crucially, personalized models can act as mechanistic classifiers (**Figure 3B**), offering insights that go beyond symptom-based categorization. By projecting empirical data onto a constrained space of neurobiological mechanisms, they enable patient stratification based on physiological parameters such as regional excitability, inhibition, or disrupted anatomical connectivity. For example, when informed by empirical priors, these models can capture how neurodegenerative processes interact with biological sex, disease progression, and regional vulnerability[99]. Simulations based on resting-state EEG and fMRI have revealed regionally imbalanced excitation-inhibition dynamics in dementia, with modulatory effects linked to sex and sociodemographic context, underscoring how structural inequalities may shape brain health trajectories[41,51,99]. Dynamical models have also been used to simulate transitions between pathological and healthy brain states, identifying control targets that could guide neuromodulation[21]. These frameworks are being extended to simulate pharmacological interventions *in silico*, integrating PET-based receptor maps to explore how neuromodulatory systems can move the dynamics in patients with disorders of consciousness closer to healthy brain dynamics[108]. Together, these approaches illustrate how WBMs can bridge neural mechanisms with contextual diversity, supporting personalized assessments and intervention strategies (**Figure 3C**).

Recent works have included HOIs in whole-brain modelling pipelines. These models have successfully been used to characterize the effects of non-invasive neuromodulation[93], and the alterations of brain dynamics in healthy aging[30]. In this way, WBMs revealed that, in aging, the increase in brain redundancy (see previous section) can be explained by a process of connectome degeneration[29]. Similar mechanisms have been proposed in neurodegeneration[22,109]. WBMs informed with metaconnectivity[110] –brain connectivity beyond pairwise interactions– also suggest that brain dynamics in Alzheimer's disease and frontotemporal dementia emerges from a combination of connectome degeneration and hypoexcitability, producing a frustrated or “viscous” brain dynamics [22]. As a limitation, usually works in the field of WMBs reproduces high-order behavior without high-order mechanisms[111,112]. A promising direction is to embed these mechanisms in WBM, i.e., by reducing the pairwise structural connectivity matrix to a small set of high-order structural connections.

Finally, brain disorders often present with overlapping clinical phenotypes despite arising from diverse biological, social, and cultural influences[17]. This phenotypic convergence shows up the limitations of one-size-fits-all explanations[113,114]. Rather than searching for universal mechanisms, it becomes necessary to account for the multiple, interacting pathways that can give rise to similar outcomes. This recognition points out the need for individualized frameworks that can identify person-specific targets and disease trajectories. WBM may tackle this challenge by offering a computational platform that integrates multimodal data to simulate brain activity at the individual level. By grounding in mechanistic principles and adapting to diverse priors, WBMs enable hypothesis testing, inference of hidden parameters, and virtual interventions tailored to each participants' brains[21,22,96,115], moving neuroscience toward a more precise, context-sensitive understanding of brain health and disease.

A synergistic computational framework for personalized medicine

Diversity, reflected in genetics, education, socioeconomic status, and environmental exposures, shapes brain health across the lifespan. Instead of filtering it out, we need approaches that remain sensitive to such variability. By combining empirical characterization with novel computational architectures, we can move beyond population averages and identify individual-level deviations that reflect both biological and contextual factors. This allows us to address diversity not as a limitation, but as a necessary dimension of precision neuroscience.

Our synergistic proposal consisted of combining all the methods presented here altogether into a diversity-sensitive integrative computational framework. HOI provide a powerful way to detect subtle and distributed alterations in brain networks, offering sensitive biomarkers even for preclinical manifestations of brain diseases[116], though this requires further exploration. This sensitivity makes HOI valuable for understanding dementia progression, tracking healthy aging, and assessing neuromodulatory interventions. By integrating HOIs into normative modeling frameworks, such as brain clocks, we can derive biologically grounded, network-specific BAGs that quantify deviations from healthy aging at the individual level. Then, WBMs can take these deviations as input, simulating the underlying biophysical dynamics and inferring causal mechanisms to better classify the different trajectories associated with individual healthy or pathological ageing. Finally, these models enable *in silico* testing of which perturbations – stimulation targets or pharmacological manipulations– most effectively reduce BAGs. This synergy transforms descriptive biomarkers into actionable mechanisms, creating a testbench for identifying potential therapeutic interventions. This synergistic loop of HOI to clocks, clocks to models, models to interventions, constitutes a computational framework with the full potential of transforming brain health and personalized medicine (**Figure 4**).

For these computational advances to have a real-world impact, they must be translated into tools suitable for clinical and community settings. Emerging technologies such as portable EEG[117], low-field MRI[118], and wearable sensors[119] offer unprecedented access to brain data outside of research laboratories. With these acquisitions setups combined with robust computational methods, even low-cost signals can be transformed into powerful indicators of individual brain health. For instance, EEG-based HOI have been proof to be sensitive enough to track the brain and subjective alterations under the effect of pharmacological interventions[120].

This approach holds promise to under-resourced settings, where access to high-end neuroimaging is limited. The goal is to create a closed-loop framework where data acquisition, model-based inference, and adaptive intervention converge, bringing precision neuroscience to the point of care.

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Author contributions

All authors equally contributed to writing and reviewing the manuscript, designing figures and boxes, and approving the final version.

Competing interests

None to declare

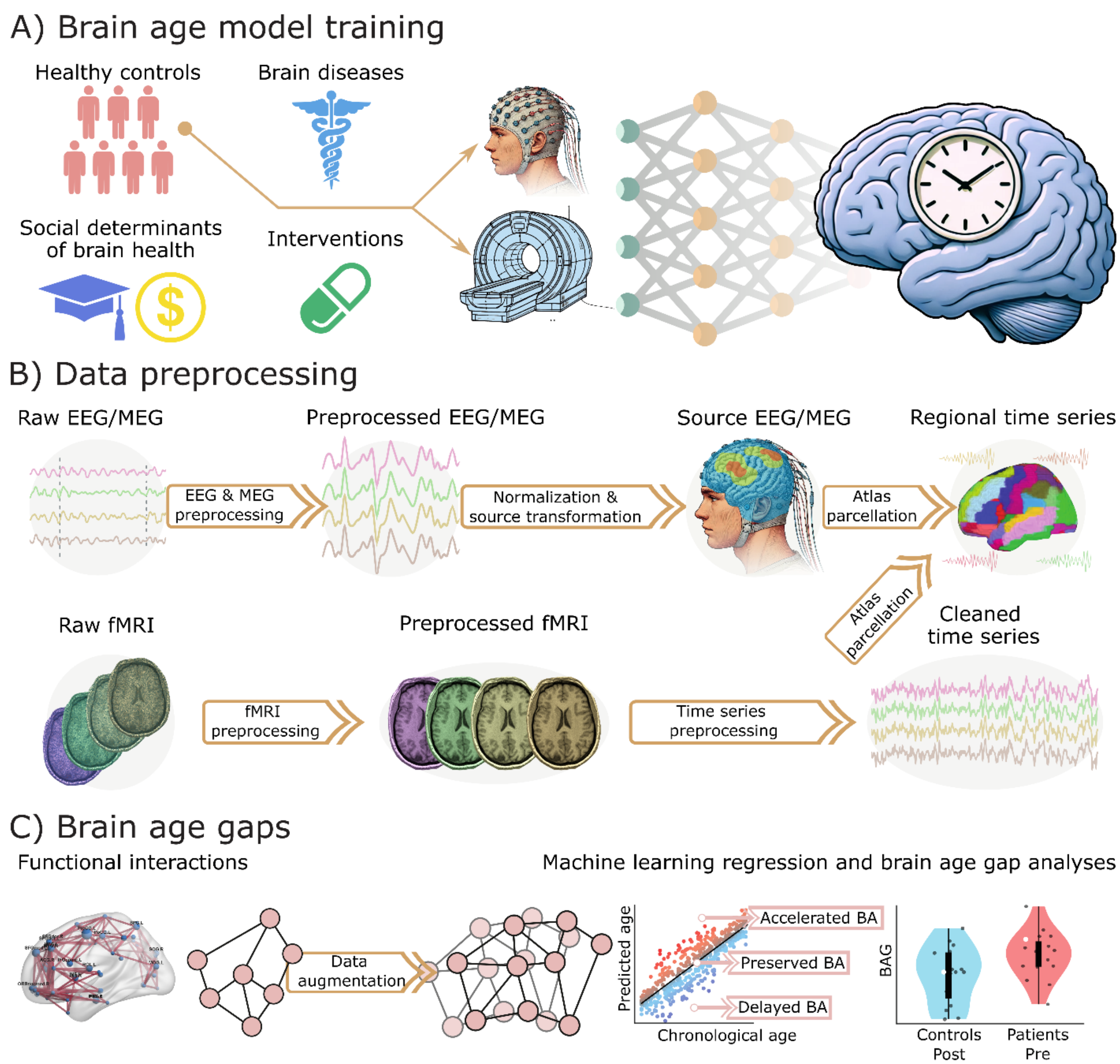

Figure 1. Functional brain clocks' architecture. A) Brain age models are trained using neuroimaging data (e.g., EEG, MEG, fMRI) from large cohorts of healthy individuals. The trained models can predict the brain age of patients with brain-related conditions and individuals exposed to varying social determinants or interventions. **B)** Data preprocessing involves signal cleaning, normalization, and transformation into a common brain space (e.g., source projection and anatomical parcellation) to extract regional time series. These features capture functional dynamics across brain networks. Non-functional data, for example, gray matter volume and white matter hyperintensities, can be used to train brain clocks or can be mixed with functional data as well. **C)** Functional connectivity matrices or graph representations are derived from the time series and used to train machine learning models (e.g., support vector machines, graph neural networks) to predict brain age. The difference between predicted and chronological age, known as the brain age gap (BAG), serves as an individualized index of brain aging. Positive BAGs suggest accelerated brain aging, while negative BAGs indicate delayed or preserved aging trajectories.

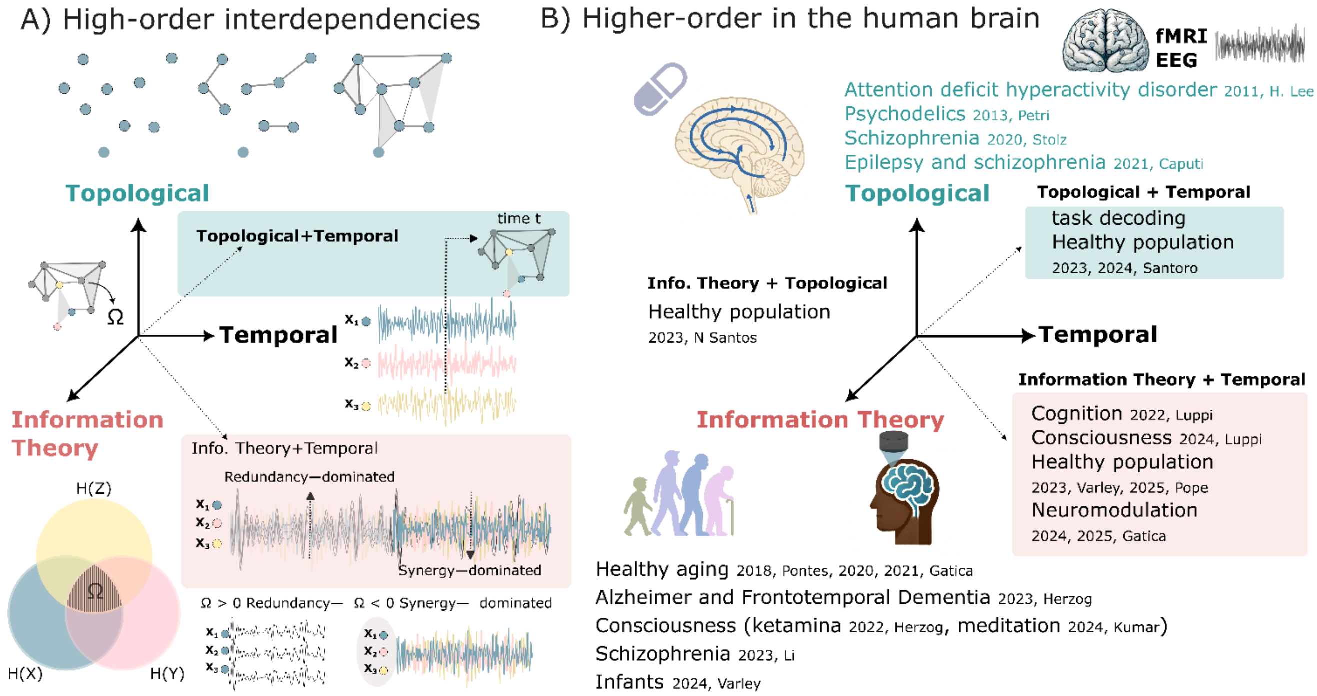

Figure 2. Topological, information-theoretic, and temporal representation of high-order interdependencies. The topological and high-order axes represent distinct analytical perspectives, while the third axis captures their time-resolved dynamics. The intersections between these axes illustrate the integrated interpretation. **A)** Conceptual overview of the three representational domains. **B)** Brain-related applications in fMRI or EEG data of healthy individuals and various neurological conditions (e.g., aging, altered consciousness, Alzheimer's disease) across the three axes.

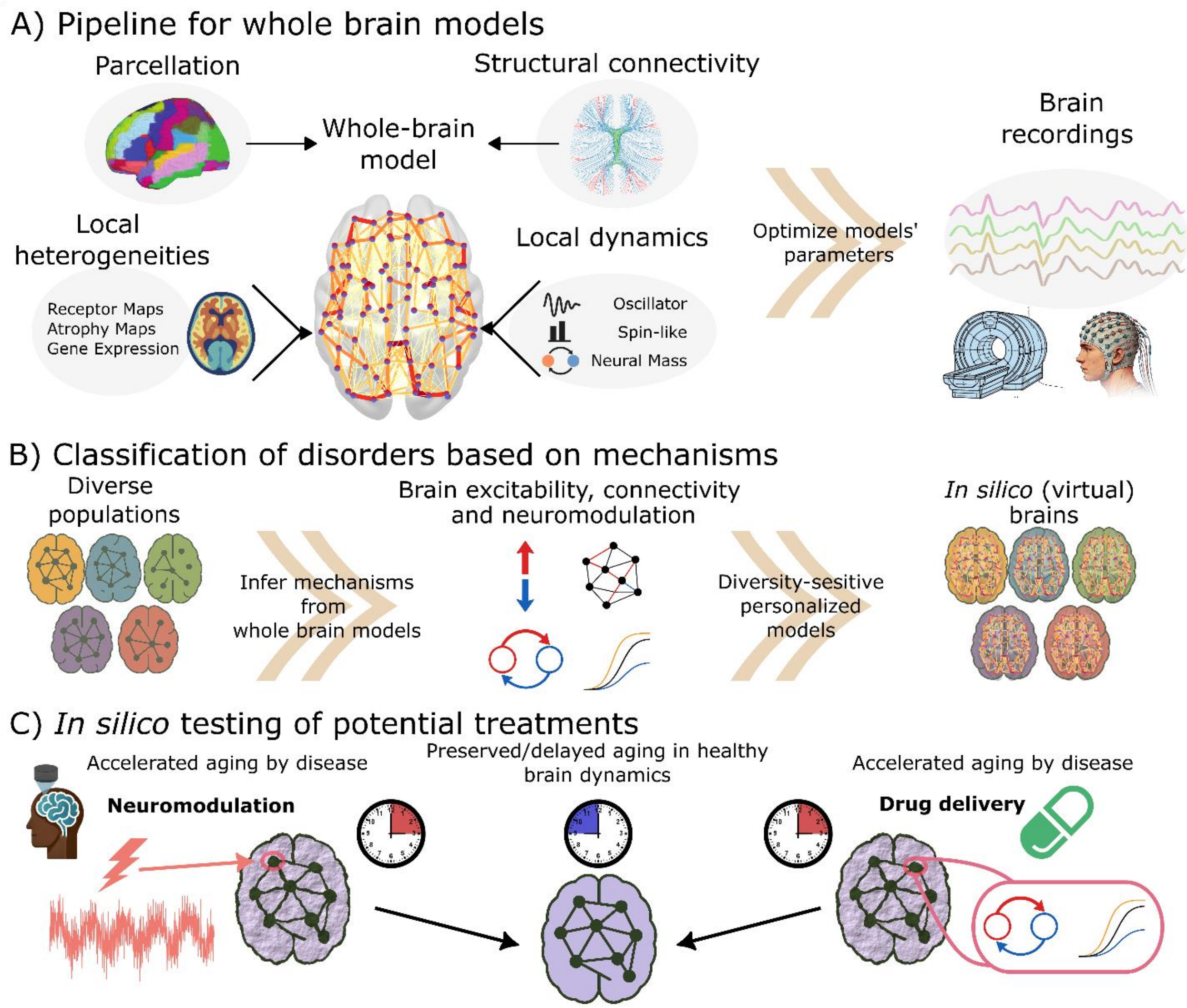

Figure 3. A framework for diversity-sensitive whole-brain modelling. A) The general pipeline for building whole brain models. Note that the local heterogeneities and the structural connectivity matrix are empirical priors, while the local dynamics depends on the level of realisms to be explained. **B)** Once whole-brain models are optimized to individual subjects or specific population, the inferred mechanisms can be used to classify different disorders. **C)** Using the diversity-sensitive whole-brain models, specific treatments can be evaluated *in silico*, accelerating the discovery of personalized treatments.

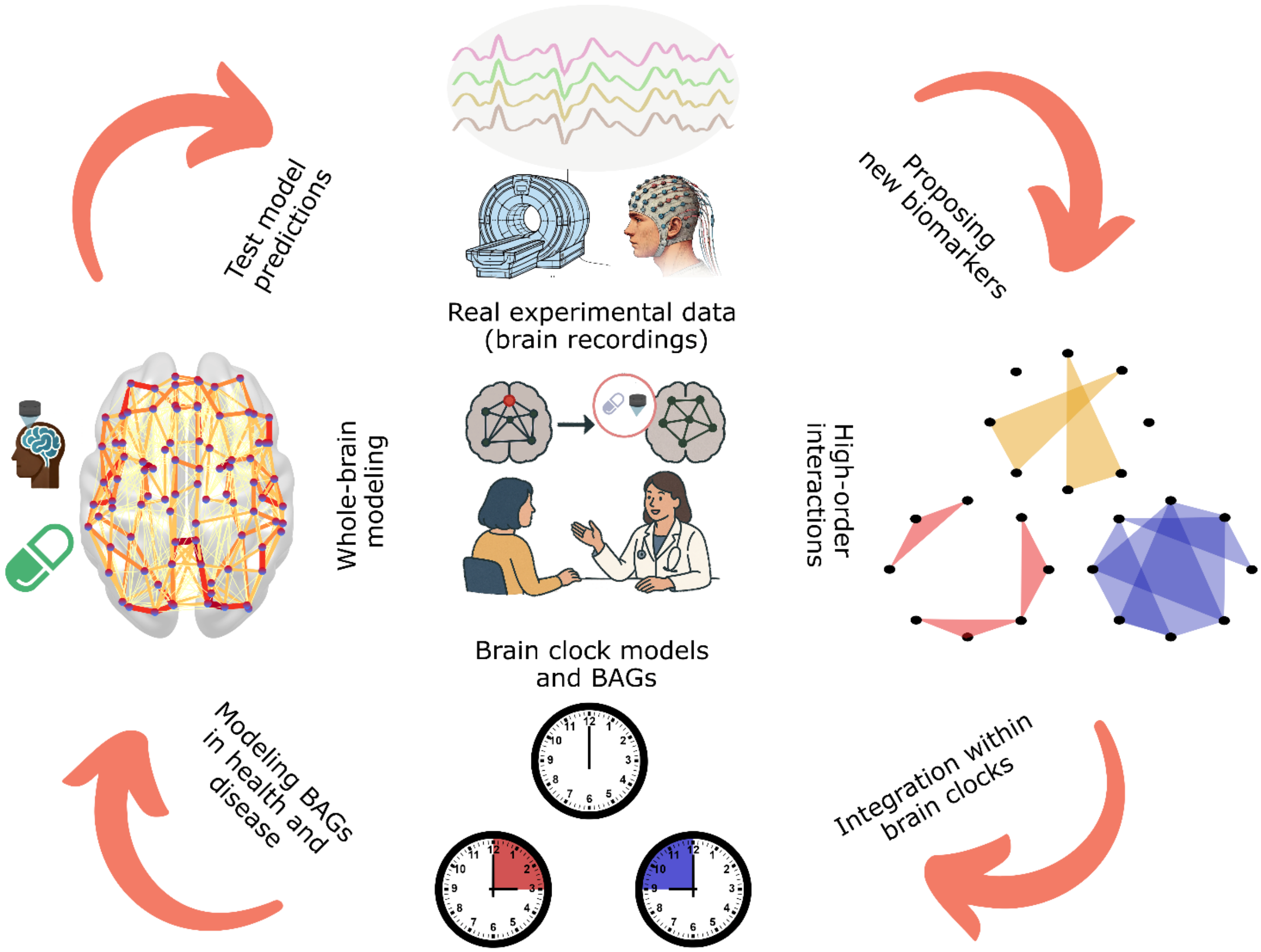

Figure 4. A synergistic computational framework for diversity sensitive personalized medicine. The figure illustrates a synergistic loop in which three complementary computational approaches (brain clocks, WBMs, and HOIs) synergize to enable a personalized characterization of brain health and disease, uncover underlying mechanisms, and inform targeted therapeutic interventions. Within this framework, brain data are first characterized using HOI analysis, providing rich, multi-scale descriptors of brain function. These descriptors can improve the estimation of biological brain age via brain clock models and the derivation of brain age gaps. Subsequently, whole-brain models provide mechanistic insights into how interventions, including pharmacological treatments, may positively influence brain health. Hypotheses generated through WBMs are then validated experimentally, requiring new data collection, closing the cycle.

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